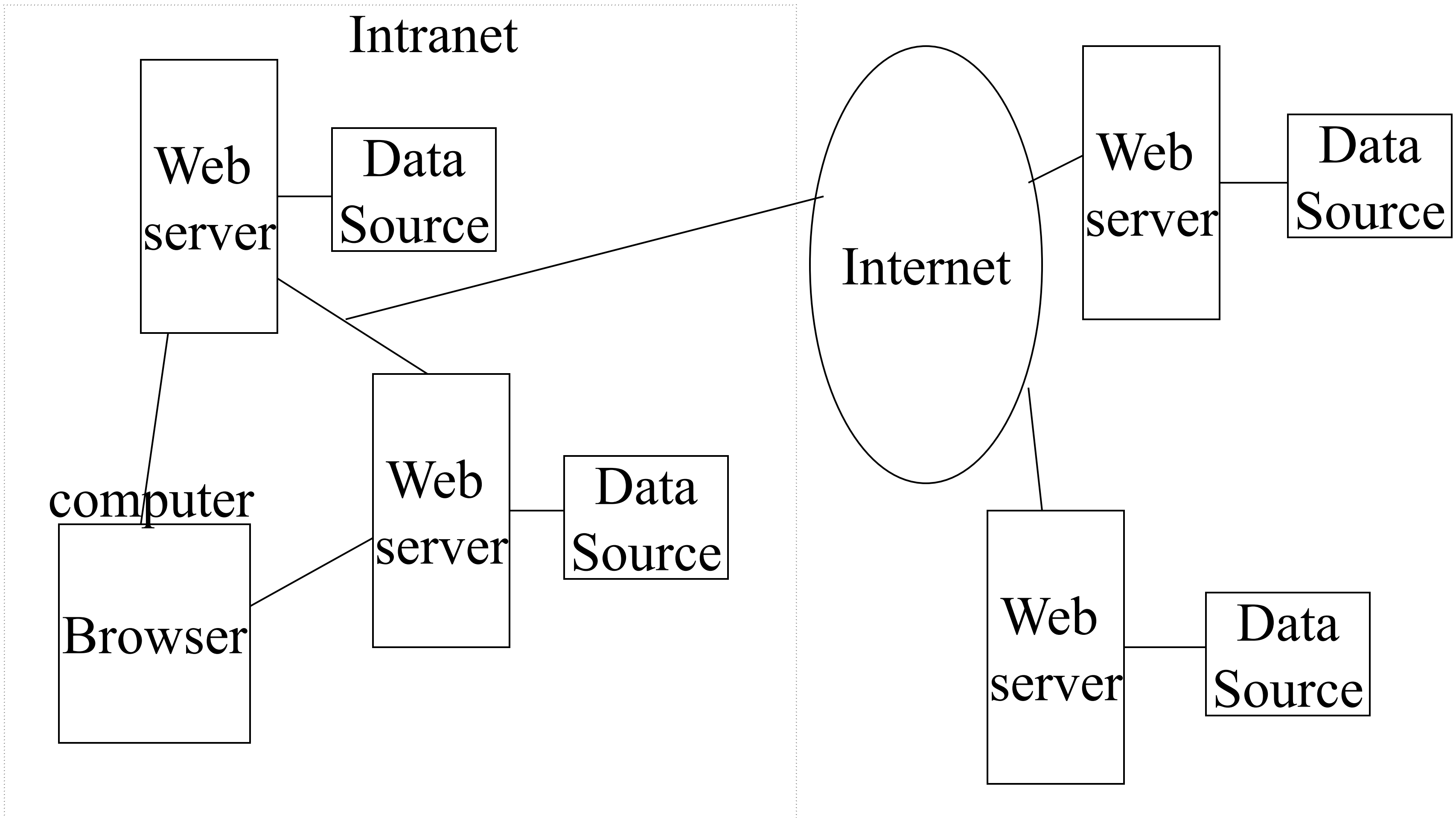
The World Wide Web (WWW)

- <u>**Material relevant to exams starts HERE with this slide.**</u>

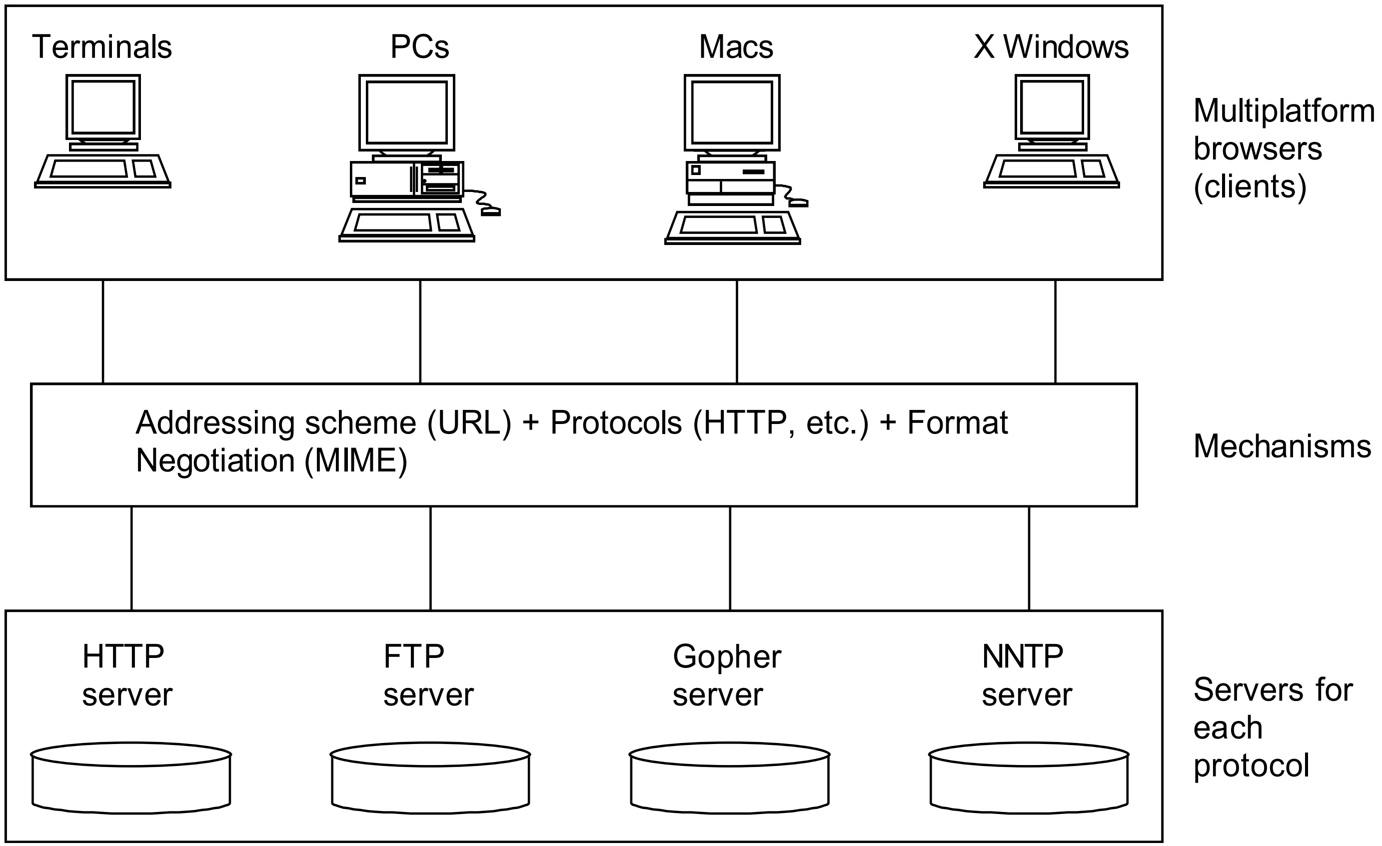
Graphical View of the WWW



Major Technology Components

- **Client/server architecture**
  - where client programs interact with web servers
- **Network protocol**
  - HTTP, Hypertext Transfer Protocol, is the language understood by browsers and web servers
  - designed to move quickly from document to document
- **Addressing system** (Uniform Resource Locators)
  - http://domain/directory/file.html
- **Markup Language**
  - every web server understands and every browser displays
  - includes support for HyperText and multimedia

Client/Server Architecture Model



The WWW Server

- Web browsers and Web servers communicate according to a protocol known as HTTP (HyperText Transfer Protocol)
  - The current HTTP protocol is version 2.0
- The Web server is a software system running on a machine often called the Web server, don't confuse them
- A web server can
  - receive and reply to HTTP requests
  - retrieve documents from specified directories
  - run programs in specified directories
  - handle limited forms of security
- A web server does not
  - know about the contents of a document, links in a document, images in a document or whether a particular file, e.g. a *.gif file, is in the correct format

Uniform Resource Locator (URL)

- A mechanism whereby an Internet resource can be specified in a single line of ASCII text
- See **RFC 1738**: http://www.faqs.org/rfcs/rfc1738.html

URL	**Refers to:**
file://pub/xt.ps	a PostScript file in directory pub on your local machine
ftp://usc.edu/docs/sweng.txt	file sweng.txt in directory docs on usc.edu, an anonymous ftp site
http://nunki.usc.edu/mydocs/book.doc	a file in directory mydocs on machine nunki.usc.edu, a WWW site
news:comp.compilers	the newsgroup computers.compilers
mailto:horowitz@usc.edu	an e-mail address

General Description of a URL

1. **Scheme** followed by a colon http:,ftp:, news:,mailto:,wais:,telnet:
2. **Double slash** (optional. Required for http, ftp) //
3. Internet **domain** name e.g., www.usc.edu
4. **Port** number (optional; e.g., www.usc.edu:8081)

   Standard or default port numbers:

---	ftp is 21	gopher is 70
---	telnet is 23	http is 80
---	smtp is 25	nntp is 119
---	imap is 143	secure nntp is 563
---	pop3 is 110	secure pop3 is 995

5. **Path** e.g., /pub/docs

URL Character Set

- RFC 1738, Dec. 1994 defines the URL character set as

"...Only alphanumerics [0-9a-zA-Z], the special characters "$-_.+!*'()," **[not including the quotes]**, and reserved characters used for their reserved purposes may be used unencoded within a URL."

- However, HTML supports ISO-8859-1 (ISO-Latin) character set
  - HTML 4.x extends the character set to all of Unicode
- Therefore, in URLs an escape mechanism is used, % followed by two hex digits
- Characters that should be encoded include:
  %, /, ., .., #, ?, ;, :, $, +, @, &, =
- Here are some encoded values for so-called “unsafe” characters

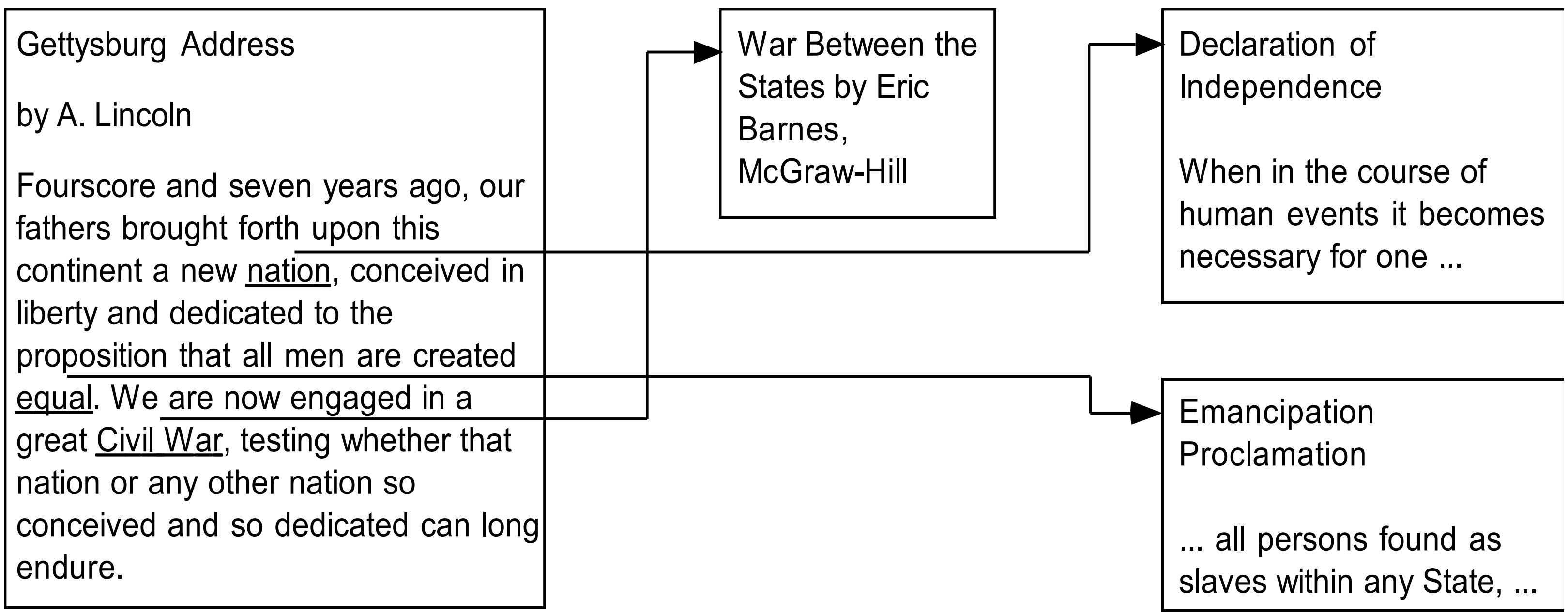
~	%7E	|	%7C
SPACE	%20	\	%5C
%	%25	^	%5E
&	%26	[	%5B
=	%3D	]	%5D
?	%3F	#	%23
{	%7B	>	%3E
}	%7D	<	%3C

Markup Languages

- HTML - hypertext markup language, specifies document layout and the specification of hypertext links to text, graphics and other types of objects
- Browsers display text and graphics using the markup as guidance
- However, HTML is *not* like a word processing program, e.g., Microsoft Word or WordPerfect, and *not* like a page description languages, e.g., postscript
  - as a result, translation into HTML can produce a result that does not look exactly like the original

What is HyperText?

- Regular text, with the additional feature of links to related documents
- As you read documents and follow links, you traverse a "web" of interconnections

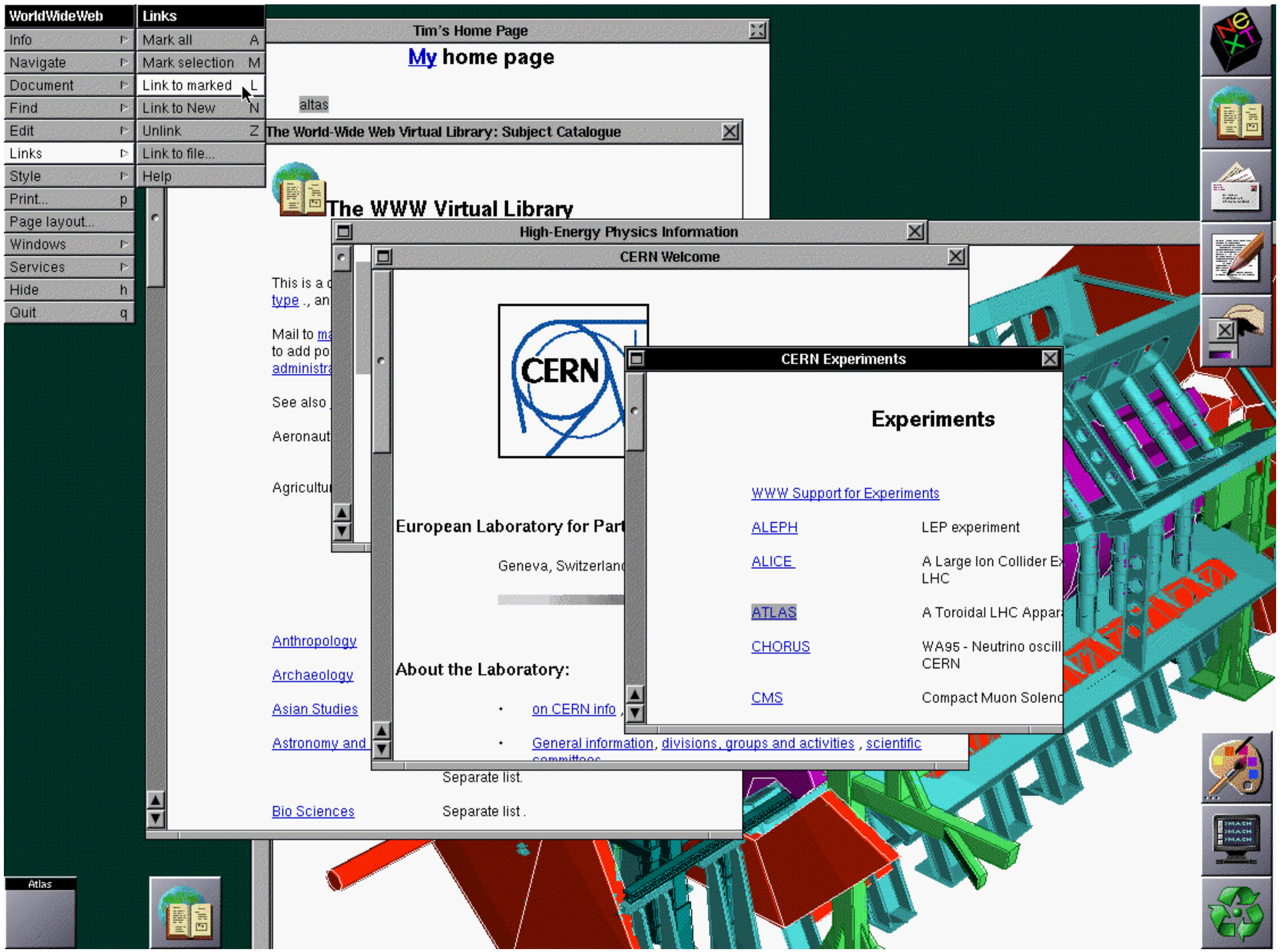


Early History of the WWW

•	1989-1990	Tim Berners-Lee conceives the WWW at CERN in Geneva
•	1990	Berners-Lee releases WWW prototype on NeXt computer
•	1992	Release of source code for line mode browser, lynx and HTTP
•	1993	Mosaic browser from NCSA is released
•	1993	WWW internet traffic now measures 1% of NSF backbone
•	12/94	Netscape Navigator 1.0 is released World Wide Web Consortium formed
•	1995	Microsoft Windows 95 and Internet Explorer 1.0 released
•	1995	Java is released
•	1998	Google is started
•	1999-2001	A burst of Internet start-up companies which flamed out because they were not profitable. Also known as he "Internet Bubble."
•	2004	Firefox 1.0 is released
•	2005	YouTube is founded
•	2008	Google Chrome 1.0 is released

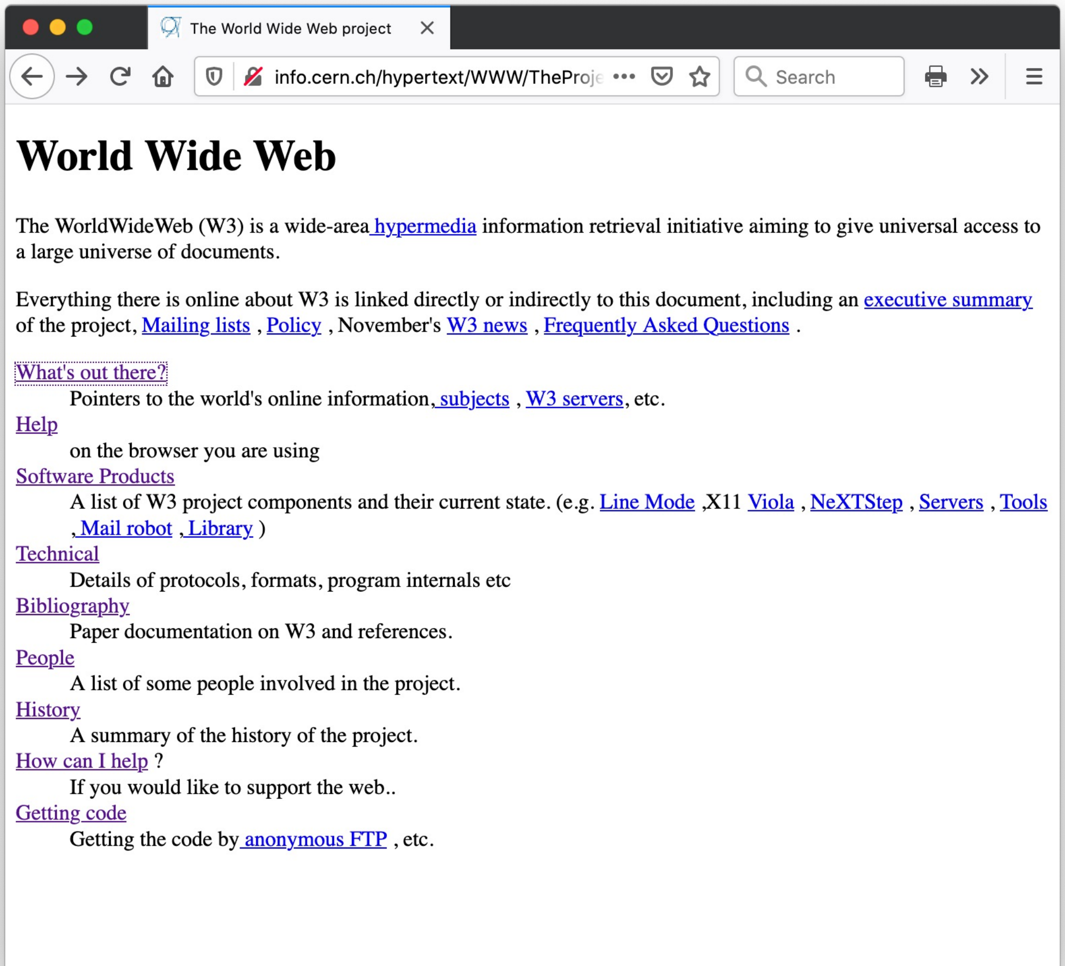
First Web Communication (Dec 1990)

See http://www.w3.org/History.html and tim Berners-Lee's presentation at the 10th anniversary, http://www.w3.org/2004/Talks/w3c10-HowItAllStarted/?n=1



Original WWW "The Project" site at CERN

http://info.cern.ch/hypertext/WWW/TheProject.html



World Wide Web

The WorldWideWeb (W3) is a wide-area hypermedia information retrieval initiative aiming to give universal access to a large universe of documents.

Everything there is online about W3 is linked directly or indirectly to this document, including an executive summary of the project, Mailing lists , Policy , November's W3 news , Frequently Asked Questions .

What's out there?
: Pointers to the world's online information, subjects , W3 servers, etc.

Help
: on the browser you are using

Software Products
: A list of W3 project components and their current state. (e.g. Line Mode ,X11 Viola , NeXTStep , Servers , Tools , Mail robot , Library)

Technical
: Details of protocols, formats, program internals etc

Bibliography
: Paper documentation on W3 and references.

People
: A list of some people involved in the project.

History
: A summary of the history of the project.

How can I help ?
: If you would like to support the web..

Getting code
: Getting the code by anonymous FTP , etc.

London Olympics (July 2012)

See http://www.zdnet.com/article/web-inventor-tim-berners-lee-stars-in-olympics-opening-ceremony/

https://www.youtube.com/watch?v=KW6ivwDcOY4

Sir Tim Berners-Lee live-tweets during the 2012 Olympics opening ceremony, with a NeXT Cube by his side

WWW Consortium

- Founded in 1994, headed by Tim Berners-Lee, http://www.w3.org
- Goal: "to lead the World Wide Web to its full potential by developing common protocols that promote its evolution and ensure its interoperability."
- Many of the technologies guided by the WWW consortium will be discussed this semester:
  - HTML, Style Sheets, Document Object Model, international character sets, HTTP, XML, etc.